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An investigation of the performance of novel chorded keyboards in combination with pointing input devices

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ABSTRACT

Rapid advances in computing power have driven the development of smaller and lighter technology products, with novel input devices constantly being produced in response to new user behaviors and usage contexts. The aim of this research was to investigate the feasibility of operating chorded keyboard control modules in concert with pointing devices such as styluses and mice. We compared combinations of two novel chorded keyboards with different pointing devices in hopes of finding a better combination for future electronic products. Twelve participants were recruited for simulation testing, and paired sample t testing was conducted to determine whether input and error rates for the novel keyboards were improved significantly over those of traditional input methods. The most efficient input device combination tested was the combination of a novel cross-shaped key keyboard and a stylus, suggesting the high potential for use of this combination with future mobile IT products.

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1. Introduction

As the mobile era has progressed, communication between people and information technology (IT) products has increasingly relied on various types of input devices. The most common tasks carried out using computer input devices include positioning, selecting, and the input of graphics and data. In general, the operation of input devices is divided into two categories according to the situations in which they are used.

First, when a computer device is attached to a secured point on a fixed workstation, the usual means of operation is the combination of a pointing device (such as a mouse) and a text input device (such as a keyboard). The pointing device performs the functions of positioning, selecting, and graphics input, while the text input device is responsible for data input functions.

Second, when traveling or in unstable operating environments, computer devices often must accommodate user movements. In such scenarios, most input operations are dependent on a pointing device (such as a stylus) to complete all the input functions. However, a single pointing device has often proven unsatisfactory to those who frequently perform text input operations. This dissatisfaction is obvious enough from the continuous new introduction of plug-in type input devices. Simply put, in many increasingly common computer operating environments, a single input device is not ideally suited to many user operations.

Many portable computer devices currently on the market contain disjointed interfaces along with incompatible input devices. It might thus be necessary for future users to learn multiple input methods, and the importance of input method integration has thus been raised previously (Isokoski and Kaki, 2000). The present research accounted for the suitability of different devices for portable computer use and previous user experiences to exclude inappropriate input devices that are currently available on the market and to devise an initial concept for combining a chorded keyboard with stylus or mouse. The design of appropriate control motions for devices and technologies must consider multiple user characteristics such as sensations, cognitions, physiological performance, and anthropometry, among others. The input requirements for mobile IT products warrant specific attention from researchers in the modern mobile era, as the extrinsic limitations of input devices often force users to learn and adapt all over again. As it is, those input devices that are not significantly improved by the incorporation of new IT product concepts will hinder users with considerable challenge in terms of input efficiency.

Accordingly, the present research consisted of a comparison of the operating conditions for different combinations of two novel chorded keyboards with two different pointing devices in hopes of finding a better combination for future electronic products.

There have been few studies focusing on chorded keyboards in the past. Initially intended for special input purposes, the chorded

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keyboard has received renewed interest since the introduction of wearable computers. Studies by Bowen and Guinness (1965) and Conrad and Longman (1965) showed that the chorded keyboards held two distinct operational advantages over conventional keyboards: smaller space requirements and higher input efficiencies. The performance of chorded keyboards is also equal to or better than that of general sequential keyboards.

The present research developed a novel chorded keyboard as primary textual and numeric input device that meets the input efficiency and size reduction requirements of modern mobile IT products.

1.1. Research purpose

The aim of this research was to investigate the feasibility of operating chorded keyboard control modules in concert with pointing devices such as styluses and mice. A study conducted by Buxton and Myers (1986) indicated that during certain operations, higher efficiency is achieved with the simultaneous use of both hands than is achieved with one hand alone. Hence, we proposed an operating condition where the left hand operates the dependent modules through a chorded keyboard, while the right hand operates the control modules and pointing functions through a stylus or mouse. We also investigated whether the proposed combinations of different keyboards with a stylus or mouse were more efficient and ergonomic than the current form of encoded design.

The burden on the left hand is expected to be reduced with an improved work balance between the two hands in terms of movement and time demands. Revealing the optimal combination form would serve as a reference for the design of future mobile IT products. Experimental comparisons between two novel chorded keyboards combined with either a stylus or mouse were conducted to determine the optimal device combination in terms of high operational performance, low error rate, and high subjective satisfaction. The four main goals of this study were: (1) to determine the operational performance and error rate of the different novel keyboard with pointing device combinations; (2) to investigate how comprehensible such combinations are by determining whether practice results in significant performance improvement; (3) to observe the operating posture of users in order to reduce possible physical injuries; and (4) to survey user satisfaction to determine advantages and disadvantages for reference in future improvements.

1.2. Literature review

1.2.1. Chorded keyboard

The existence of keyboards dates back over a hundred years ago. Computer keyboards are the primary input devices for data entry operations, and the most commonly used keyboard among visual display terminal (VDT) operators is the conventional QWERTY keyboard, which is used by almost all computer users. Chorded keyboards, in contrast, use considerably fewer keys than conventional keyboards to produce similar inputs. The operation of chorded keyboards involves simultaneously pressing two to three keys in order to enter a single alphabetic or numeric character. For example, with one previously tested chorded keyboard, five keys could be used to produce 31 chord arrangements to represent alphabets, numbers, phrases, commands, or other strings (Noyes, 1983), and for some handicapped people, chorded keyboards have proven to be the only suitable type of keyboard for regular use. A study by Buxton (2002) suggested that the input method of chorded keyboards holds several advantages:

1. Chorded keyboards have fewer keys and are smaller in size than conventional keyboards, which make them far more suitable for portable use (Cumming, 1984).
2. Chorded keyboards support one-handed operation, which is beneficial to handicapped users (Kirschenbaum et al., 1986; Heusinkveld, 1988).
3. Chorded keyboards can be operated when the use is mobile or in an otherwise unstable environment.
4. The use of chorded keyboards results in reduced muscle fatigue and injuries in comparison to conventional keyboard use (Fisher and Bond, 1992).

In comparison to sequential keyboards, the speed (key strokes per minute, KSPM) of chorded keyboard operation is slower than that of sequential keyboards, but chorded keyboards provide higher data input volumes. In general, chorded keyboards are

Table 1
Analysis of one-hand and two-handed keyboards.

	Product	Factors
One-handed keyboards	Stealthy keyboard	The Stealthy keyboard proposed by Mckown was based on chorded keyboards (Mckown, 2000). The six buttons produce combinations to accommodate 36 characters, and functional buttons that switch input between alphabets and symbols are included The users need to learn different combinations for input, and adapt to new input methods
	Wrist PC keyboard	A small QWERTY keyboard designed to fit on the wrist (L3 system). The hand wearing Wrist PC keyboard holds the mobile device while the other hand performs one-handed input The one-handed input method and the small buttons lead to poor input efficiency
Two-handed keyboards	Mehring	Input function is achieved through contact of different conductors (Mehring, 2003). The conductors are distributed at the phalangeal joints and phalangeal regions to allow alphabetic or numeric input by combinations of rows and columns. It is difficult for the distal phalanges of the little finger and the ring finger to make contact with the three joints of the thumb. Additional training of user is required because of different special symbol distribution to normal keyboards; The relative positioning of keys and the large amount of conductors make the manufacturing of electric circuits difficult.
	AG-5	An input device developed based on the arcade machine joystick (Martinez et al., 2005). A trackball is also installed into the device that makes the AG-5 perform both the functions of keyboard and mouse. In terms of the keyboard, AG-f integrated buttons for independent keys and multiple valued keys. The most used alphabets are placed at the posterior for Touch-typing input, while the rest of the alphabets are arranged on the anterior. The anterior keys are not arranged according to typing layouts. In terms of operation, fast switching of fingers between the anterior and posterior is difficult. The employment of multiple valued keys also makes a poor symbol input performance by AG-5.

difficult to learn, particularly the associated encoding systems. However, decent input performance can be achieved after familiarization.

Previously, one-handed chorded keyboards were common among physically handicapped users because they can help the handicapped as long as they hit a large button or input a chord (Kirschenbaum et al., 1986); these keyboards have also grown increasingly popular among users who require fast input and easy portability (McMulkin and Kroemer, 1994). Recent advances in chorded keyboard designs have resolved ergonomic issues and reduced keyboard areas to improve their feasibility for broader use. Table 1 provides a brief analysis of two types of one-handed keyboards and two types of two-handed keyboards, while Table 2 presents a comparison of different key-related input methods.

The cross-shaped key keyboard (Fig. 1) and the square quad-character keyboard (Fig. 2) employed in this study needed to be combined with pointing devices and act as one-hand operated keyboards similar to the Wrist PC keyboard. Hence, ergonomic adjustments were made to the size of the keys for accurate target key stroke by the fingers, and the key spacing was also measured for adjustments to lower the rate of inadvertent strokes. A diagram of both types of chorded keyboards used in this study combined with pointing input devices is presented in Fig. 3.

1.2.2. Input methods

The input methods of current mobile IT products are abundant, which could be roughly divided into the following types. Single pointing device: input is performed with a single pointing device, which is common for products such as PDA; plug-in keyboards: an input device to mitigate the inadequacy of a single pointing device; embedded keyboard: keyboard is reduced and embedded into the product (Tablet PC). For example, the IPAD device designed and manufactured by APPLE can be seen as a new input device with an embedded keyboard and reduced in size.

Input devices overly reduced in size led to higher fatigues (Hahn, 2002), and the use of multiple character keys reduces the area of the keyboard without reducing the area of the keys. This research adopted the familiar QWERTY keyboard as reference to the key order layout to facilitate fast learning of chord combinations.

2. Material and methods

2.1. Novel input device user evaluation

2.1.1. Purpose

Input devices of well-known brands that are currently available in the market were investigated for simulation experiment. These products were used to study the characteristics of various models in the market to understand the factors why keyboard input devices and pointing input devices are unergonomic. Then, the design parameters of the novel input devices were incorporated to observe and assess their applicability, and revised in detail according to the characteristics of the products to achieve user convenience.

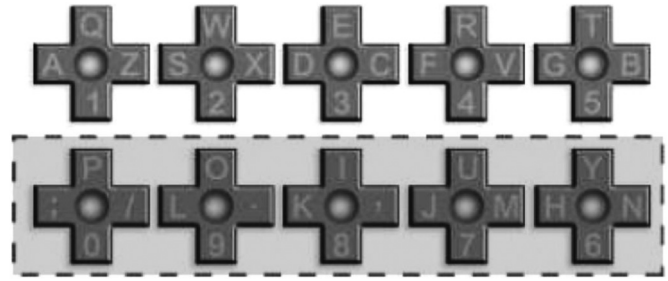

Fig. 1. Concept diagram of cross-shaped, simulated Braille keys.

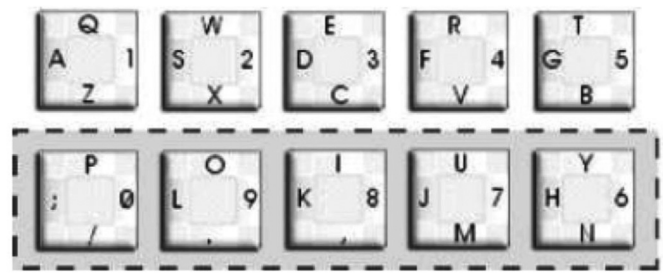

Fig. 2. Concept diagram of square, quad-character, simulated Braille keys.

2.1.2. Test subjects

Twelve test subjects consisting of males and females (50% each) that were recruited for this study aged between 20 and 30 years, frequently use technology products, are right-handed and do not have any severe disabilities, have over 3 years of experience in using computer input devices (including pointing input and text input devices), and who use related products more than 5 times a week.

2.1.3. Tools

Tools used in this study included digital video cameras, electromyography, model making tools, motion capture system, Rapid Simulation software, and Flash.

2.1.4. Method

To verify the effect novel input device designs have on input devices of information products today, actual simulation test and

Table 2
A comparison between different input methods.

Category	Advantages	Disadvantages	Applicable scope
Every key is limited to only one function	• Easy to distinguish • Easy to operate • Clearly visible function	• Require large keyboard area • Unsuitable for large data entry	• Repetitive use of functions • Significant functional difference • Difficult to break into groups
Every key represents a few functions	• Reduced number of keys • Standard keyboard can be used	• High occurrence of error • Operation often require stroking two or more keys • Distribution is difficult to optimize	• Functions can be grouped and easy to learn
Non-cornered keys that represents only two character functions	• Small keyboard size and production miniaturization • Conventional keyboard layout with raised familiarity	• Difficult to learn without typing training • Keys covering too many characters could be confusing	• Dual control modules of keys in combination with pointing • Functions can be grouped for easy learning

Adapted from Preece et al., 1994.

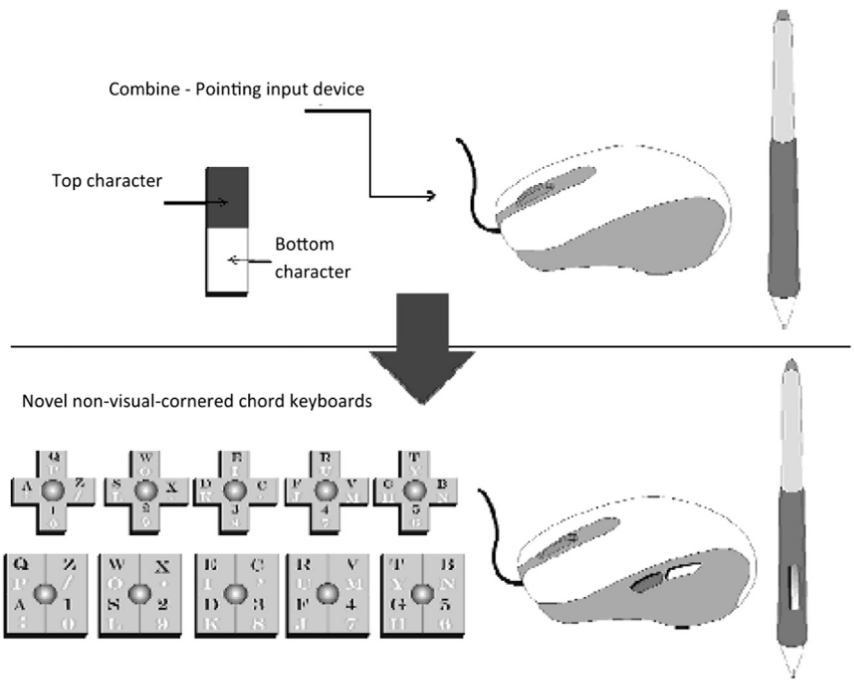

Fig. 3. Schematic diagram of simulated Braille keyboards combined with pointing input devices.

users' subjective questionnaire method were conducted to see if there are any differences between users using the novel input device of this study and traditional input method. To ensure objective measurements of the subjects' performance rates, the subjects were pre-trained before testing so that each would reach a plateau on his or her learning curve, thus preventing errors due to mere interpersonal differences between the subjects. In addition, parameters were revised to obtain information products that are commonly used in the current market.

The input validity of concept products with novel input devices and related products that belong to the same market was compared. In addition, the validity, physiological load, error rate, and other usages were also evaluated. After the experiment, the test subjects were asked to fill out a user evaluation questionnaire to obtain the actual results of applying novel input devices on information products.

2.2. Concept development

The combination of novel non-cornered chords accounted for the impact of ergonomic key positioning which effectively reduces the possible physical injuries from input operations. The consideration of ergonomic principles was made according to previous related literature investigations (Saito et al., 2000; Nakaseko et al., 1985). The principles of an ergonomic keyboard are summarized into five groups: (1) the operating principle of keyboard columns; (2) the principle of two hands operation; (3) the principle of hands burden; (4) the principle of clustering; and (5) the principle of array.

2.3. Concept presentation

The primary goal of this research was to compare the usage performance of two novel chorded keyboards and also to compare the usage performance of the same chorded keyboards in combination with different pointing devices. The optimal chord arrangement is to be determined through revelation of the input device set with the best input rate and accuracy and specific assessment of user physiological and mental sensations by a subjective questionnaire.

2.4. Experimental parameters

2.4.1. Independent variables

The chord combinations of the two chorded keyboard sets were chosen based on their common usage and previous research. Tests were conducted with four sets of devices (A1, A2, B1, and B2), including combinations with a stylus or mouse. The combination sets are shown visually in Fig. 4 and specified in table form in Table 3.

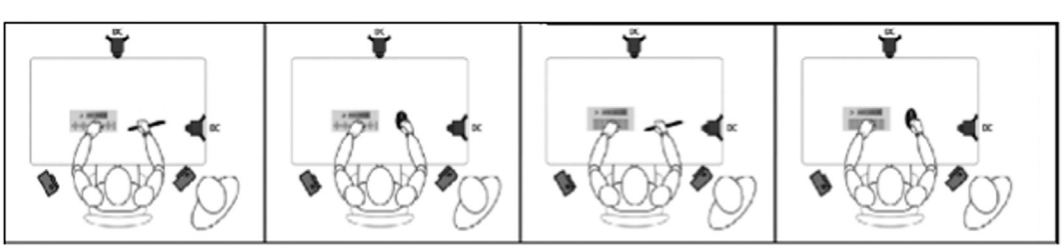

Fig. 4. Schematic representation of experiment combination sets.

Table 3
Paired sample *T*-test of the four chord device sets input rate from day 1 and day 3.

	Difference of paired variables				*t* value	Degrees of freedom	Significance (two-tail)
	Mean	Standard deviation	95% Confidence interval of difference				
			Upper limit	Lower limit			
Cross-shaped directional keyboard with dual control module stylus	−83.4	48.72	−118.25	−48.55	−5.41	9	.000[a]
Cross-shaped directional keyboard with dual control module mouse	−91.6	56.40	−131.94	−51.26	−5.14	9	.001[a]
Square cornered chorded keyboard with dual control module stylus	−66.9	22.69	−83.13	−50.67	−9.32	9	.000[a]
Square cornered chorded keyboard with dual control module mouse	−76.2	35.53	−101.61	−50.79	−6.78	9	.000[a]

[a] Indicates statistical significance $p < 0.05$.

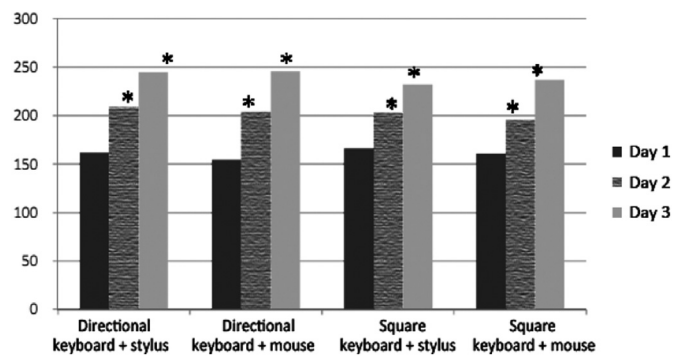

Fig. 5. Graph of text input rate analysis.

2.4.2. Dependent variables

(1) Input rate: the number of characters entered according to the articles provided by researcher within the experiment time (six minutes)
(2) Error rate: the number of incorrect characters divided by the total number of entered characters
(3) Subjective questionnaire

2.4.3. Control variables

(1) Test subjects
(2) Operation related variables: novel keyboard groupings and two pointing device character entry
(3) Environmental variables: the environment was kept constant for each experiment, including the lighting, furnishing, and configuration of the experiment venue.

2.5. Experimental process and procedures

Input performance data of the different chorded keyboard sets were collected and values of time spent and incorrect character entered by the subjects were recorded. The subjects completed the subjective questionnaire at the end of each experiment to express their mental and physiological sensations. The first stage of the experiment process was primarily input method familiarization and practice, and the experimental data collection was performed at the second stage. The experiment included textual and numeric entry, drag, click, and selection of generic document processing. Prior to the training session, each user must familiarize themselves with the chord input method and the key positions on the keyboard. A code table was provided to help the subjects understand each chord arrangement, and every subject was allowed fifteen minutes of practice prior to the six minute experiment. Typing of the English articles prepared in advance was performed during the experiment; character entry was completed by chorded keyboard in combination with the stylus or mouse. Every set of experiment was six minutes long. A five minute practice period was allowed prior to the start of experiment, and a three minute break was provided between the practice and experiment period. The subjects performed the aforementioned experimental procedures for three days and completed the subjective questionnaire at the end of the third day to complete the process.

3. Results and discussion

3.1. Analysis of chord arrangement text input rate improvement

The first stage of experimental analysis was the statistical analysis of the input rates of the four chord device sets. Paired sample *T*-test was performed on data collected on day 1 and day 3 to determine whether the chord device input rate of users had significantly improved after three days of training. As shown in Fig. 5, the input rate at day 3 was significantly improved in comparison to the rate of day 1, indicating that the input rate of a user can be improved through training. In addition, the input rate improved in a linear fashion and the magnitude of improvement gradually decreased with the accumulation of training days. The improvement rate from day 1 to day 3 was the greatest recorded.

Table 4
Paired sample *T*-test of the four chord device sets input error rates of day 1 and day 3.

	Difference of paired variables				*t* value	Degrees of freedom	Significance (two-tail)
	Mean	Standard deviation	95% Confidence interval of difference				
			Upper limit	Lower limit			
Cross-shaped directional keyboard with dual control module stylus	1.863	2.44	0.12	3.61	2.42	9	.039[a]
Cross-shaped directional keyboard with dual control module mouse	2.249	2.22	0.66	3.84	3.20	9	.011[a]
Square cornered chorded keyboard with dual control module stylus	1.216	0.47	0.88	1.55	8.22	9	.000[a]
Square cornered chorded keyboard with dual control module mouse	2.073	2.11	0.56	3.59	3.10	9	.013[a]

[a] Indicates statistical significance $p < 0.05$.

Table 5
Subjective assessment analysis.

Overall assessment of the device sets of two chorded keyboards with two pointing devices	(cross-shaped directional keyboard with dual control module stylus) > (cross-shaped directional keyboard with dual control module mouse) > (square cornered keyboard with dual control module stylus) > (square cornered keyboard with dual control module mouse)
Comfort of chorded keyboard operation	Cross-shaped directional keyboard > square cornered keyboard
Comfort of pointing device operation	Stylus > mouse
Cognitive learning of chorded keyboards	Cross-shaped directional keyboard > square cornered keyboard
Cognitive learning of pointing device	Mouse > stylus

Table 4 presents statistical data for the four different chord device set input rates over the period from day 1 to day 3.

3.2. Analysis of chord arrangement text input error rate improvement

The second stage of experimental analysis was the statistical analysis of the input error rates of the four chord device sets. Paired sample *T*-tests were performed on the data collected on day 1 and day 3 to determine whether the chord device input error rate had significantly improved after five days of user training. Table 5 indicates that the input error rate recorded on day 3 significantly improved from that of day 1, which suggests that the input error rate can be lowered by user training. Other than *T*-tests, a repeated measures approach would also be another appropriate way to perform statistics analyses.

Additionally, as shown in Fig. 6, the improvement in the input error rate did not keep to an elevating trend as the number of experimental days increased. The error rates of the four device sets displayed a trend of gradual decrease with the increase in training days. While a relatively large improvement of the input rates between day 1 and day 3 was observed, such a large improvement was not observed for the input error rate.

3.3. Subjective questionnaire analysis

The users were required to complete a subjective questionnaire after the experiment ended. The questionnaire adopted a non-segmented Likert scale of 1–10, and the users scored freely on the scale to provide consecutive values. The content of questionnaire focused on investigating the difficulty of experimental input device operation perceived by the test subjects (with the values of 1 and 10 representing easiest and most difficult, respectively) and hand comfort during experiment (with the values of 1 and 10 representing most comfortable and most uncomfortable, respectively). The overall questionnaire result for the keyboard showed that subjects felt the keyboard size was too large and hence needs improvement. As the current keyboard could only perform input functions and not simultaneously perform input, selection, and movement function, a hand needs to be freed to control a subsequent command. The direction of research was highly rated.

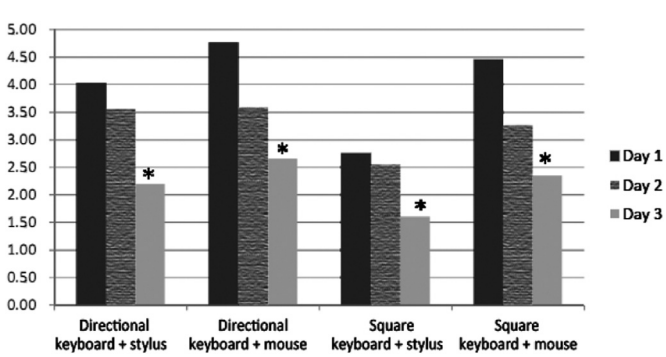

Fig. 6. Graph of text input error rate analysis.

3.4. Subjective comfort satisfaction comparison result

Operation of the first device set (cross-shaped directional keyboard with dual control module stylus) was rated as most comfortable on the left hand with a mean score of 6.07. However, the overall comfort assessment differed with the decrease in dependent modules of the keyboard, where the hand felt more comfortable as the number of operated dependent modules decreased. In terms of right hand comfort, operation of the third device set (square cornered keyboard with dual control module stylus) was rated as the most comfortable with a mean score of 5.63. It was observed that the comfort of right hand was influenced by the operation of different pointing devices, and it appears that the operation of stylus pointing device was more comfortable than operation of mouse pointing device.

3.5. Subjective learning satisfaction comparison result

The learning performance of three training days was poorest for the third device set (square cornered keyboard with dual control module stylus) with a mean score of 7.84; the second device (cross-shaped directional keyboard with dual control module mouse) was rated as the easiest to learn, with a mean score of 7.38. The learning ability of test subjects was influenced by the different combinations of device sets where those that included cross-shaped directional keyboard were easier to learn and more recognized by the subjects, which also made the set easier to adopt by the subjects.

After three days of input training, it was discovered that the operation difficulty of the two chorded keyboards with two pointing devices perceived by the test subjects had all improved.

An explanation for the lower visual recognition of square cornered keyboard than that of cross-shaped key keyboard is that the directional nature of the cross-shaped key keyboard better prevented confusion of operation. There was less difference between the square cornered keyboard and the current keyboard, but the relatively dense arrangement of alphabets was difficult to search by users. A bulk of the subjects suggested that future development should combine non-cornered keyboard design with stylus and touch screen to achieve simultaneous text entry and perform one-handed operation of displacement commands.

4. Conclusions

The rapid development of technology has led people living in the information era to embrace new technologies and enjoy the resulting convenience. Previous studies have found that approximately 80% of people are exposed to IT products in their daily lives. Mobile IT products have now become a part of daily life for many people, and it is necessary to develop novel input methods specifically for IT products.

It was suggested by previous research that working with both hands brought a higher operating efficiency than working with just one hand (Damper, 1986). Novel combinations of stylus with chord entry were proposed by this research and compared to the entry methods of current products, and the best combination was determined. The results could serve as a reference to the design of future input devices for table computers.

The conclusion and recommendations of this study are as follows:

1. Input efficiency of chorded keyboard in combination with stylus and mouse pointing devices gradually improved with training – this study conducted device set building with two novel chorded keyboards with stylus and mouse pointing devices. The user input efficiency with all devices sets improved with the increase in usage days. The input rate of chorded keyboard significantly improved after short period of training, and a decent input rate was achieved.
2. We compared the character string sets input rate of device set combinations of cross-shaped directional and square cornered chorded keyboards with pointing devices of stylus and mouse. Encoding method that focused on character string input allowed users to quickly achieve particular input rate. However, a shortcoming of earlier arrival of input rate at the convergence value than other methods was also observed. After short periods of training the test subjects effectively recalled the chord arrangements from memory, and continuous improvement was possible.
3. New combinations of chords created from the current chord arrangements were helpful to chord arrangements familiarization and effective learning by the test subjects. Therefore, variations of the current chords could be included in the development of chords for touch-controlled tablet computer keyboards to encourage users to embrace the chords within a short period of time.
4. The utilization of a stylus as pointing device for mobile communication products effectively reduces injuries associated with certain operating posture, while maintaining an input rate similar to that of a mouse. The best input device combination observed in this study is the combination of a novel cross-shaped key keyboard and a stylus, which fulfilled the input rate requirement of mobile IT products.

Acknowledgments

The manuscript has been carefully edited and reviewed by an experienced editor, Mr. Daniel Boles, whose first language is English and who specializes in editing papers written by scientists whose native language is not English.

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